

Math 161

Problem set 5 solutions

Justin Campbell

1. (a) Suppose that $\lim_{x \rightarrow 0^+} f(x) = L$, so we would like to show that $\lim_{x \rightarrow 0} f(|x|) = L$, i.e. for any $\epsilon > 0$ there exists $\delta > 0$ such that for any $x \in \mathbb{R}$, the inequality $0 < |x| < \delta$ implies that $|f(|x|) - L| < \epsilon$. Since $\lim_{x \rightarrow 0^+} f(x) = L$ by assumption, there exists $\delta > 0$ such that for any $x' \in \mathbb{R}$, the inequality $0 < x' < \delta$ implies that $|f(x') - L| < \epsilon$. Taking $x' = |x|$, it follows that $0 < |x| < \delta$ implies that $|f(|x|) - L| < \epsilon$ as needed.

Conversely, suppose that $\lim_{x \rightarrow 0} f(|x|) = L$. To see that $\lim_{x \rightarrow 0^+} f(x) = L$, we must prove that for any $\epsilon > 0$, there exists $\delta > 0$ such that for any $x \in \mathbb{R}$, the inequality $0 < x < \delta$ implies that $|f(x) - L| < \epsilon$. By our hypothesis $\lim_{x \rightarrow 0} f(|x|) = L$, there exists $\delta > 0$ such that $0 < |x| < \delta$ implies that $|f(|x|) - L| < \epsilon$. If $0 < x < \delta$, then in particular $x = |x|$ and $0 < |x| < \delta$, so we obtain

$$|f(x) - L| = |f(|x|) - L| < \epsilon$$

as desired.

- (b) First assume that $\lim_{x \rightarrow a} f(x) = L$, so we would like to show that $\lim_{x \rightarrow a^+} f(x) = L$ and $\lim_{x \rightarrow a^-} f(x) = L$, i.e. for any $\epsilon > 0$ there exist $\delta_1, \delta_2 > 0$ such that for any $x \in \mathbb{R}$, the inequality $0 < x - a < \delta_1$ implies that $|f(x) - L| < \epsilon$, and $-\delta_2 < x - a < 0$ also implies that $|f(x) - L| < \epsilon$. Since $\lim_{x \rightarrow a} f(x) = L$, there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies that $|f(x) - L| < \epsilon$. Taking $\delta_1 = \delta_2 = \delta$, we see that if $0 < x - a < \delta$, then in particular $x - a = |x - a|$ and hence $0 < |x - a| < \delta$, so $|f(x) - L| < \epsilon$. Similarly, if $-\delta < x - a < 0$ then $|x - a| = -(x - a)$ and hence $0 < |x - a| < \delta$, which again implies that $|f(x) - L| < \epsilon$.

For the converse, assume that

$$\lim_{x \rightarrow a^+} f(x) = L = \lim_{x \rightarrow a^-} f(x).$$

We would like to show that $\lim_{x \rightarrow a} f(x) = L$, which means that for any $\epsilon > 0$ there exists $\delta > 0$ such that for any $x \in \mathbb{R}$, the inequality $0 < |x - a| < \delta$ implies that $|f(x) - L| < \epsilon$. Since $\lim_{x \rightarrow a^+} f(x) = L$, there exists $\delta_1 > 0$ such that $0 < x - a < \delta_1$ implies that $|f(x) - L| < \epsilon$. Similarly, we have $\lim_{x \rightarrow a^-} f(x) = L$ and hence there exists $\delta_2 > 0$ such that $-\delta_2 < x - a < 0$ implies that $|f(x) - L| < \epsilon$. Let $\delta := \min(\delta_1, \delta_2)$. If $0 < |x - a| < \delta$, then either $0 < x - a < \delta$ or $-\delta < x - a < 0$. In the first case we have $0 < x - a < \delta \leq \delta_1$ and hence $|f(x) - L| < \epsilon$. In the second case we have $-\delta_2 \leq -\delta < x - a < 0$, which also implies that $|f(x) - L| < \epsilon$.

- (c) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by

$$f(x) := \begin{cases} 1 & \text{if } x \geq 0, \\ -1 & \text{if } x < 0. \end{cases}$$

Then we have

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 1 = 1$$

because $f(x) = 1$ for $x > 0$, and

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} -1 = -1$$

because $f(x) = -1$ for $x < 0$. Since $\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$, the limit $\lim_{x \rightarrow 0} f(x)$ does not exist by part (b).

2. (a) We must show that for any $M > 0$, there exists $\delta > 0$ such that $0 < |x| < \delta$ implies that $\frac{1}{x^2} > M$. So take $\delta := \min(1, \frac{1}{M})$. Then $0 < |x| < \delta \leq 1$ implies that $|x|^2 < |x|$. Since $x^2 = |x|^2$, we therefore have $x^2 < |x|$. On the other hand, we have

$$0 < x^2 < |x| < \delta \leq \frac{1}{M},$$

from which it follows that $\frac{1}{x^2} > M$ as desired.

- (b) The statement that $\lim_{x \rightarrow 0} \frac{1}{x} = \infty$ is false means that there exists $M > 0$ such that for any $\delta > 0$, there exists $x \in \mathbb{R}$ such that $0 < |x| < \delta$ and $f(x) \leq M$. So take $M := 1$. For any $\delta > 0$, let x be any number satisfying $-\delta < x < 0$, e.g. $x = -\frac{\delta}{2}$. Then we have $0 < |x| < \delta$ and

$$f(x) = \frac{1}{x} < 0 < 1 = M.$$

3. (a) First suppose that $\lim_{x \rightarrow \infty} f(x) = L$, so we want to show that $\lim_{x \rightarrow 0^+} f(\frac{1}{x}) = L$, i.e. for any $\epsilon > 0$ there exists $\delta > 0$ such that for any $x \in \mathbb{R}$, the inequality $0 < x < \delta$ implies that $|f(\frac{1}{x}) - L| < \epsilon$. Our assumption that $\lim_{x \rightarrow \infty} f(x) = L$ implies that there exists $M > 0$ such that for any $x' \in \mathbb{R}$ satisfying $x' > M$, we have $|f(x') - L| < \epsilon$. Take $\delta := \frac{1}{M}$, so if $0 < x < \delta$ then $\frac{1}{x} > M$, which implies that $|f(\frac{1}{x}) - L| < \epsilon$ as needed.

For the converse, assume that $\lim_{x \rightarrow 0^+} f(\frac{1}{x}) = L$. We would like to prove that $\lim_{x \rightarrow \infty} f(x) = L$, i.e. for any $M > 0$ there exists $\delta > 0$ such that for any $x \in \mathbb{R}$, the inequality $x > M$ implies that $|f(x) - L| < \epsilon$. Our hypothesis $\lim_{x \rightarrow 0^+} f(\frac{1}{x}) = L$ implies that there exists $\delta > 0$ such that for any $x' \in \mathbb{R}$, the inequality $0 < x' < \delta$ implies that $|f(\frac{1}{x'}) - L| < \epsilon$. Put $M := \frac{1}{\delta}$. Then $x > M$ implies that $0 < \frac{1}{x} < \delta$, whence

$$|f(\frac{1}{1/x}) - L| = |f(x) - L| < \epsilon.$$

- (b) Note that since $\lim_{x \rightarrow \infty} f(x)$ only depends on $f(x)$ for $x > 0$, and

$$\frac{2x^2 + x - 5}{3x^2 + 7} = \frac{2x^2 + x - 5}{3x^2 + 7} \cdot \frac{1/x^2}{1/x^2} = \frac{2 + 1/x - 5/x^2}{3 + 7/x^2}$$

for $x \neq 0$, we have

$$\lim_{x \rightarrow \infty} \frac{2x^2 + x - 5}{3x^2 + 7} = \lim_{x \rightarrow \infty} \frac{2 + 1/x - 5/x^2}{3 + 7/x^2}.$$

Now apply part (a), which tells us that

$$\lim_{x \rightarrow \infty} \frac{2 + 1/x - 5/x^2}{3 + 7/x^2} = \lim_{x \rightarrow 0^+} \frac{2 + x - 5x^2}{3 + 7x^2}.$$

In fact, we claim that

$$\lim_{x \rightarrow 0} \frac{2 + x - 5x^2}{3 + 7x^2} = \frac{2}{3},$$

which implies that the right-handed limit also equals $\frac{2}{3}$ by 1(b). To see this, recall from lecture that rational functions are continuous at any point where they are defined. Since $3 + 7 \cdot 0^2 = 3 \neq 0$, we obtain

$$\lim_{x \rightarrow 0} \frac{2 + x - 5x^2}{3 + 7x^2} = \frac{2 + 0 - 5 \cdot 0^2}{3 + 7 \cdot 0^2} = \frac{2}{3}.$$

4. (a) We will need the inequality

$$||x| - |y|| \leq |x - y|,$$

which holds for all $x, y \in \mathbb{R}$. We already know that $|x| - |y| \leq |x - y|$, so it suffices to observe that

$$-|x - y| = -|y - x| \leq -(|y| - |x|) = |x| - |y|.$$

Assuming that f is continuous at a , we will show that $|f|$ is continuous at a . The latter means that for any ϵ , there exists $\delta > 0$ such that $|x - a| < \delta$ implies that $||f|(x) - |f|(a)| < \epsilon$. Since f is continuous at a , there exists $\delta > 0$ such that $|x - a| < \delta$ implies that $|f(x) - f(a)| < \delta$. But then $|x - a| < \delta$ implies that

$$||f|(x) - |f|(a)| = ||f(x)| - |f(a)|| \leq |f(x) - f(a)| < \epsilon.$$

- (b) Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) := \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ -1 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Then $|f|$ is the constant function 1 and hence continuous everywhere. On the other hand, we claim that f is continuous nowhere, i.e. for any $a \in \mathbb{R}$ there exists $\epsilon > 0$ such that for any $\delta > 0$, there exists $x \in \mathbb{R}$ such that $|x - a| < \delta$ and $|f(x) - f(a)| \geq \epsilon$. Take $\epsilon := 1$, and choose x so that $a - \delta < x < a + \delta$ and

- $x \in \mathbb{Q}$ if $a \notin \mathbb{Q}$;
- $x \notin \mathbb{Q}$ if $a \in \mathbb{Q}$.

This is always possible because both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$. Then by our choice of x we have $|x - a| < \delta$, and

$$|f(x) - f(a)| = 2 > 1 = \epsilon$$

because either $f(x) = 1$ and $f(a) = -1$, or $f(x) = -1$ and $f(a) = 1$.

5. Suppose that $\lim_{x \rightarrow a} f(x) = L$, and assume also that $U \subset \mathbb{R}$ is open and $L \in U$. We must produce an open set $V \subset \mathbb{R}$ satisfying the three desired properties. Since U is open and $L \in U$, there exists $\epsilon > 0$ such that $|y - L| < \epsilon$ implies that $y \in U$. Now use the hypothesis that $\lim_{x \rightarrow a} f(x) = L$, so there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies that $x \in A$ and $|f(x) - L| < \epsilon$. Define $V := (a - \delta, a + \delta)$, so certainly V is open and $a \in V$. If $x \in V \setminus \{a\}$, then $0 < |x - a| < \delta$ and hence $x \in A$, which shows that $V \setminus \{a\} \subset A$. Moreover, the inequality $0 < |x - a| < \delta$ implies that $|f(x) - L| < \epsilon$, which by our choice of ϵ implies that $f(x) \in U$. Thus $f(V \setminus \{a\}) \subset U$.

Now we prove the converse. We must show that for any $\epsilon > 0$, there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies that $|f(x) - L| < \epsilon$. Let $U := (L - \epsilon, L + \epsilon)$. Since U is open and $L \in U$, by hypothesis there exists an open set $V \subset \mathbb{R}$ such that $a \in V$, $V \setminus \{a\} \subset A$, and $f(V \setminus \{a\}) \subset U$. The fact that U is open and $a \in V$ implies that there exists $\delta > 0$ such that $(a - \delta, a + \delta) \subset V$. If $x \in \mathbb{R}$ satisfies $0 < |x - a| < \delta$, then

$$x \in (a - \delta, a + \delta) \setminus \{a\} \subset V \setminus \{a\} \subset A,$$

and moreover $x \in V \setminus \{a\}$ implies that

$$f(x) \in f(V \setminus \{a\}) \subset U = (L - \epsilon, L + \epsilon).$$

The latter means that $|f(x) - L| < \epsilon$ as desired.